INVENTORY OPTIMIZATION AND ADVANCED ANALYSIS SYSTEM

Submitted By:

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Course :

SAP ABAP Developer

Project Title:

Inventory Optimization and Advanced Inventory Analysis using ABAP

Institution:

Kalinga Institution of Industrial Technology

Date of Submission: 21/04/2026

2. Problem Statement

Inventory management is a crucial aspect of any organization dealing with physical goods. Inefficient inventory handling leads to several issues, such as:

- Excess stock (overstocking), resulting in increased holding costs
- Shortage of stock (understocking), leading to lost sales opportunities
- Lack of visibility into demand patterns
- Absence of decision-support systems for inventory planning

Traditional systems mainly provide raw stock data but fail to deliver meaningful insights like:

- Which items need immediate replenishment
- Which items are overstocked or obsolete
- What actions should be taken to optimize inventory

Therefore, there is a need for a system that not only tracks inventory but also analyzes it, classifies it, and provides actionable recommendations.

3. Solution / Features

This project introduces a two-level inventory management system:

3.1 Basic Inventory Optimization

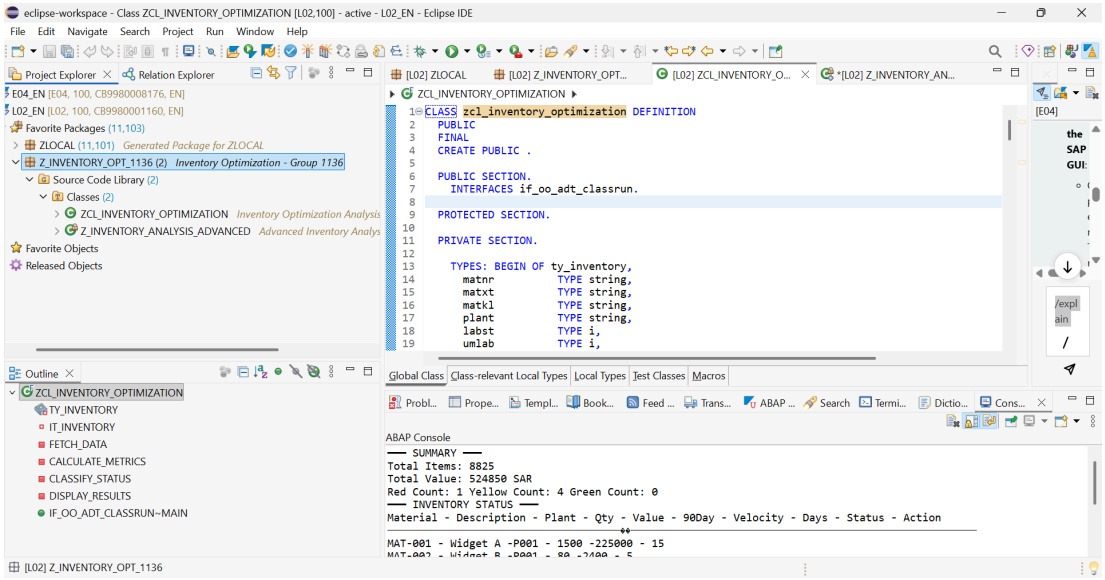
This module focuses on fundamental inventory evaluation.

Key Functionalities:

- Fetches inventory data (material, plant, quantity, price)
- Calculates:
    - **Stock Value** = Quantity × Price
    - **Velocity** (FAST / SLOW) based on movement
    - **Days of Supply** based on consumption rate
- Classifies inventory into:
    - **RED** → Critical stock
    - **YELLOW** → Warning
    - **GREEN** → Healthy
- Provides recommendations such as:

- Urgent restocking
- Monitoring stock
- Maintaining current levels

(Class: `ZCL_INVENTORY_OPTIMIZATION`)



Purpose:

To provide a quick overview of inventory health and basic decision-making support.

3.2 Advanced Inventory Analysis

This module enhances the system with deeper analytics and intelligent insights.

Key Functionalities:

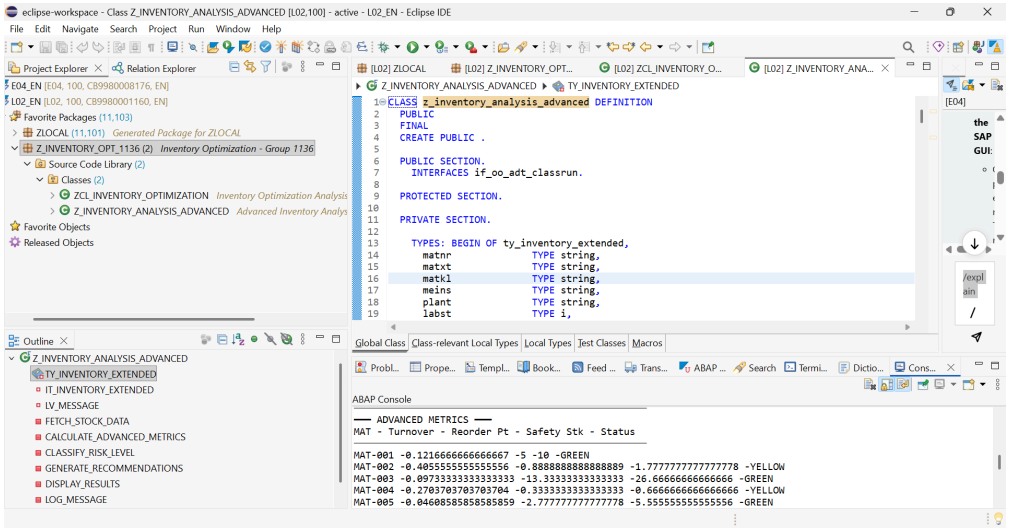
- Tracks material movement over:
  - 90 days
  - 30 days
  - 7 days
- Calculates advanced metrics:
  - **Stock Value**
  - **Turnover Ratio**
  - **Days of Supply**
  - **Reorder Point**
  - **Safety Stock**

- Classifies velocity into:
  - VERY_FAST, FAST, NORMAL, SLOW, VERY_SLOW
- Determines risk levels:
  - **CRITICAL** → No stock
  - **HIGH** → Likely stockout
  - **MEDIUM** → Needs attention
  - **OVERSTOCK** → Excess inventory
  - **OBSOLESCENCE** → Very low movement
  - **LOW** → Healthy
- Generates intelligent recommendations:
  - Emergency ordering
  - Scheduled replenishment
  - Stock reduction strategies
  - Product phase-out decisions
- Assigns priority levels:
  - P0 → Immediate
  - P1 → Today
  - P2 → This week
  - P3 → Review
  - P4 → Monitor

Purpose:

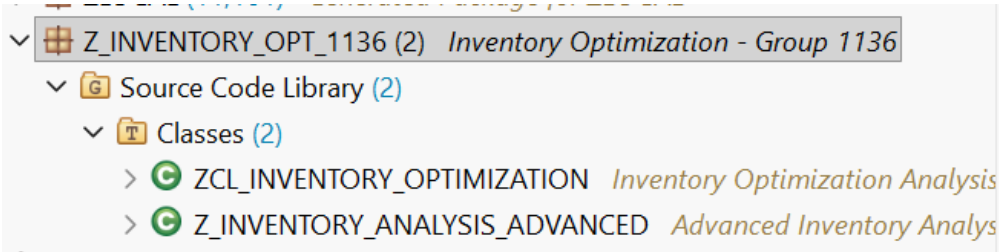
To provide detailed insights and support strategic inventory decisions.

(Class: `Z_INVENTORY_ANALYSIS_ADVANCED`)



4. Screenshots

4.1 Project Structure



4.2 Output Display (Execution Result)

Basic Class `ZCL_INVENTORY_OPTIMIZATION` (RESULTS)

```
——— SUMMARY ———
Total Items: 8825
Total Value: 524850 SAR
Red: 1 Yellow: 4 Green: 0

——— INVENTORY STATUS ———

MAT ID    | Description | Plant |   Qty    | Value   | Days | Velocity | Status     | Action
-------------|----------------|---------|----------|----------|-------|-----------|-------------|--------------------
MAT-001 | Widget A     | P001  | 1500   | 225000 | 999  | FAST    | YELLOW | OPTIMIZE Overstock
MAT-002 | Widget B     | P001  | 80       | 2400     | 999  | SLOW   | YELLOW | CAUTION Monitor
MAT-003 | Widget C     | P002  | 5000   | 150000 | 999  | FAST    | YELLOW | OPTIMIZE Overstock
MAT-004 | Widget D     | P001  | 45       | 2250     | 999  | SLOW   | RED        | CRITICAL Low Stock
MAT-005 | Widget E     | P002  | 2200   | 145200 | 999  | SLOW   | YELLOW | REVIEW High Stock
-------------|----------------|---------|----------|----------|-------|-----------|-------------|--------------------
```

The system's results indicate that the inventory is not properly balanced. Although the total stock quantity (8825 units) and total value (524850 SAR) are high, the distribution of inventory across materials is uneven. A significant portion of the stock is concentrated in a few items, indicating that capital is not being utilized efficiently.

In the basic inventory optimization output, most items are classified as YELLOW, with only one item marked as RED and none as GREEN. This means that there is no material in an ideal or healthy condition. The RED item (MAT-004) represents a critical low stock situation, where immediate replenishment is required. On the other hand, several items have high stock levels but low consumption, which results in inefficient

inventory holding. The very high “days of supply” values (999) further indicate that stock will last for a long time due to low usage, pointing toward overstocking.

Advanced class `Z_INVENTORY_ANALYSIS_ADVANCED` (RESULTS)

```
╠═══════════════════════════════════════════════════
  ADVANCED INVENTORY ANALYSIS RESULTS
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——— SUMMARY STATISTICS ———
Total Materials: 6
Total Stock Qty: 8825
Total Stock Value: 524850 SAR
Critical: 1 High: 0 Medium: 2 Overstock: 0
Obsolete: 0 Healthy: 3
——— DETAILED ANALYSIS ———
MAT - Description - Plant - Qty - Value - Days - Velocity - Risk - Priority
———————————————————————————————————————————
MAT-001 -Widget A Premium -P001 -1500 -225000 -3000 -FAST -LOW -P4 MONITOR
MAT-002 -Widget B Standard -P001 -80 -2400 -900 -SLOW -MEDIUM -P2 THIS WEEK
MAT-003 -Widget C Industrial -P002 -5000 -150000 -3750.000000000001 -VERY_FAST -LOW -P4
MONITOR
MAT-004 -Widget D Economy -P001 -45 -2250 -1350 -VERY_SLOW -MEDIUM -P2 THIS WEEK
MAT-005 -Widget E Heavy -P002 -2200 -145200 -7919.999999999999 -NORMAL -LOW -P4 MONITOR
MAT-006 -Widget F Specialty -P001 -0 -0 -0 -NORMAL -CRITICAL -P0 IMMEDIATE
———————————————————————————————————————————
———————————————————————————————————————————
——— ADVANCED METRICS ———
MAT - Turnover - Reorder Pt - Safety Stk - Status
———————————————————————————————————————————
MAT-001 -0.1216666666666667 -5 -10 -GREEN
MAT-002 -0.4055555555555556 -0.8888888888888889 -1.7777777777777778 -YELLOW
MAT-003 -0.09733333333333333 -13.33333333333333 -26.66666666666666 -GREEN
MAT-004 -0.2703703703703704 -0.3333333333333333 -0.6666666666666666 -YELLOW
MAT-005 -0.04608585858585859 -2.777777777777778 -5.555555555555556 -GREEN
MAT-006 -0 -1.666666666666667 -3.333333333333334 -RED
———————————————————————————————————————————
——— RECOMMENDATIONS ———
MAT - Risk Level - Recommendation - Priority
———————————————————————————————————————————
MAT-001 -LOW -Maintain current stock levels All good -P4 MONITOR
MAT-002 -MEDIUM -Schedule order for next purchase cycle -P2 THIS WEEK
MAT-003 -LOW -Maintain current stock levels All good -P4 MONITOR
MAT-004 -MEDIUM -Schedule order for next purchase cycle -P2 THIS WEEK
MAT-005 -LOW -Maintain current stock levels All good -P4 MONITOR
MAT-006 -CRITICAL -URGENT Stock depleted Emergency order required -P0 IMMEDIATE
———————————————————————————————————————————
```

The advanced inventory analysis provides deeper insights into the situation. One material (MAT-006) is classified as CRITICAL, meaning it is completely out of stock but still has demand. This is the most serious issue, as it can directly impact operations or sales. Two materials fall under the MEDIUM risk category, indicating that their stock levels are not immediately critical but still require attention and timely replenishment.

The turnover ratio is generally low, which means inventory is not circulating efficiently. This reflects poor utilization of resources, as money is tied up in stock that is not being used or sold quickly. Finally, the recommendation system correctly prioritizes actions. The critical item is assigned immediate priority, medium-risk items are scheduled for near-term action, and low-risk items are marked for monitoring. This shows that the system is effectively converting analysis into actionable decisions.

Overall, the results clearly indicate that the inventory system suffers from both overstocking and understocking, along with inefficient stock utilization. The advanced analysis successfully identifies these issues and provides a structured approach to improve inventory management.

5. Tech Stack

- **Programming Language:** ABAP (Object-Oriented ABAP)
- **Platform:** ABAP Cloud
- **Development Tool:** Eclipse IDE with ABAP Development Tools (ADT)

6. Unique Points

- Combines basic and advanced inventory analytics in one system
- Implements real-world inventory KPIs used in industries
- Provides risk-based classification instead of simple thresholds
- Generates actionable recommendations automatically
- Introduces priority-based decision making (P0–P4)
- Simulates intelligent decision logic using ABAP
- Modular and scalable design using Object-Oriented principles

7. Future Improvements

- Integration with SAP standard tables:
    - MARA (Material Master)
    - MARD (Storage Location Data
- Real-time data instead of static values
- Development of a SAP Fiori dashboard for visualization
- Integration with Machine Learning models for demand forecasting
- Automatic generation of Purchase Orders (PO)

- Integration with SAP MM (Material Management module)

8. Conclusion

This project demonstrates how ABAP can be used not only for data processing but also for intelligent business decision-making. By combining inventory tracking with analytical logic, the system helps organizations:

- Reduce costs
- Avoid stockouts
- Improve operational efficiency

The advanced module further enhances the system by providing deep insights and strategic recommendations, making it a practical and scalable solution for real-world inventory management.